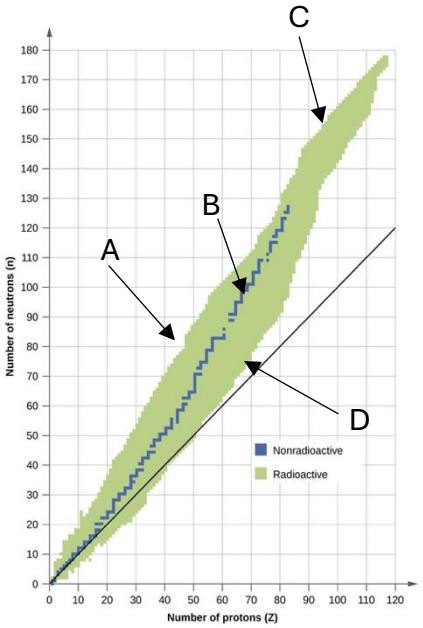
Nuclear Unit Homework #1 Name: _____

1. In this unit, we are focusing on processes that occur in what part of the atom? _____
2. Explain how a nuclear process differs from a chemical reaction? What particles are usually involved in a chemical reaction?

3. Explain what forces are occurring in the nucleus that lead to stability and instability.

4. An isotope that is considered not radioactive: _____
5. Every element past _____ contains no isotopes that are stable.
6. Elements that are radioactive will undergo what process and why?

7. The graph to the right shows the band of stability for known isotopes of every element. Match each arrow with the descriptions below.



____ The isotope is proton rich and will want to change a proton to a neutron

___ The isotope is too heavy to be stable and will want to lose protons and neutrons

___ The proton to neutron ratio is stable and the isotope is not radioactive.

___ The isotope is neutron rich and will want to change a neutron to proton.

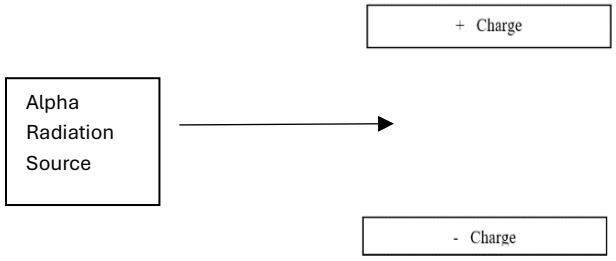
There is an interactive band of stability website that is linked in Schoology.

Nuclear Homework #2 Name: _____

We have learned that an unstable nucleus will want to undergo radioactive decay to become more stable.

1) **What are the 3 main types of radioactive decay** we have discussed and **how does each type change the nucleus?**

2) A students sets up an experiment where they use shoot alpha particles through an electromagnetic field created by 2 magnets. On the diagram below, draw how the alpha particles behave in the magnetic field and write a short explanation why.



3) A radioactive source that produces alpha particles is only dangerous if somehow ingested. Alpha particles carry a lot of energy and can ionize many atoms in your cells. Why are alpha particles not dangerous outside your body?

4) Beta radiation, beta minus specifically, can penetrate through your skin. What material can absorb beta radiation?

5) Many nuclear reactions can produce gamma radiation which is extremely dangerous because it passes through everything. What properties of gamma radiation allows for this and what kind of material do you need to use to absorb gamma radiation?

6) For each problem below, write the decay equation(s) for the radioactive isotope.
   a. Cl-34 is proton rich and will undergo beta plus decay

   b. Mn-54 is neutron rich and will undergo beta minus decay

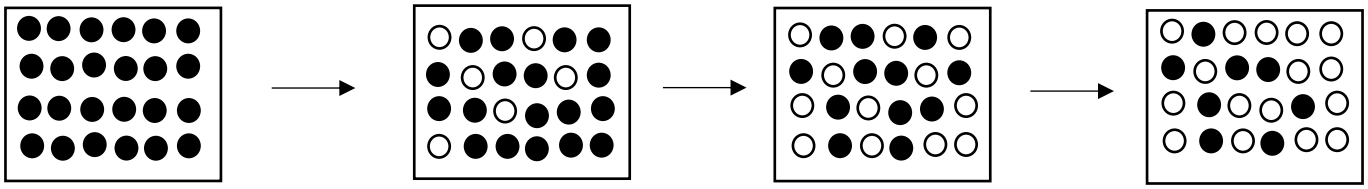
   c. Rn-216 will undergo two successive alpha decays to become a stable isotope.

Nuclear Homework #3 Name: _____

1. What is the half-life of a substance?

2. A radioactive isotope has a half-life of 25 seconds. How many half-lives would 200.0 grams of that substance need to undergo to only have 12.5 grams remain? How many seconds would it take for that change to occur?

3. The following diagrams show the change of element A (solid circle) into element B . There are 24 atoms total of element A at a time of 0 seconds.

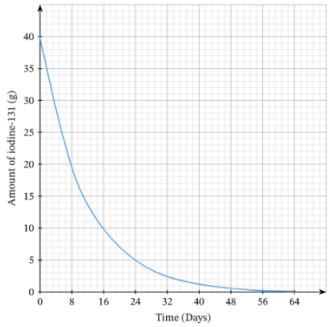


a. What is the half-life of element A? Justify your answer using the data provided in the diagrams above.

b. Draw the diagram for the sample if a total of 60 seconds has passed since the first diagram.

4. Use the graph provided to the right to answer the questions below.
   a. What is the half-life for Iodine-131?

   b. How many grams of iodine are left after it undergoes 3 half-lives?



5. What is the half-life of an isotope that has an initial mass of 100 grams and a final mass of 3.13 grams after 30 days?

6. An isotope has a half-life of 20 seconds. How many seconds have passed if the isotope decreases from 75.0 grams to 9.38 grams?

Nuclear Homework #4

1. Both nuclear fission and nuclear fusion involve changes to the nucleus which release energy. Briefly describe how each of the processes change the nucleus. Talk about what types of isotopes undergo these processes.

2. Nuclear fusion requires EXTREME temperature and pressure to occur. Explain why these conditions are necessary for nuclear fusion to occur. Think about the forces that exist inside the nucleus and why it would be hard to fuse one nucleus to another nucleus.

3. In a nuclear fission reaction, explain how the fission produces a chain reaction. Be specific by talking about what is produced in the reaction and how that continues the process.

4. Balance the following fission/fusion reactions. Remember that the products of the process need to balance with the reactants!

$^{231}_{91}Pa \rightarrow {}^{192}_{77}Ir + _____$

Fission or fusion

$^{9}_{4}Be + {}^{12}_{6}C \rightarrow _____$

Fission or fusion

$^{244}_{94}Pu \rightarrow {}^{141}_{59}Pr + _____$

Fission or fusion

$^{238}_{92}U \rightarrow {}^{128}_{52}Te + _____$

Fission or fusion

$^{4}_{2}He + {}^{2}_{1}H \rightarrow _____$

Fission or fusion

$^{56}_{26}Fe + {}^{59}_{28}Ni \rightarrow _____$

Fission or fusion

$^{262}_{105}Db \rightarrow {}^{96}_{42}Mo + {}^{28}_{14}Si + _____$

Fission or fusion

$^{265}_{108}Hs \rightarrow {}^{16}_{8}O + {}^{9}_{4}Be + _____$

Fission or fusion